

WORKSHEET 1

- Workers in a construction may go on strike. The probability that the workers go on strike is 0.60. The probability of completing the reconstruction on the date is 0.35 if workers go on strike, 0.85 if they do not go. Find the probability that the reconstruction will be completed on the date.

Solution:

$$\frac{60}{100} \cdot \frac{35}{100} + \frac{40}{100} \cdot \frac{85}{100} = \frac{21}{100} + \frac{34}{100} = \frac{55}{100} = 0.55$$

- There are 15 blue balls and 8 red balls in the box A, 19 blue balls and 15 red balls in the box B, 12 blue balls and 18 red balls in the box C. A ball is randomly selected from a random box. If it is seen that the selected ball is blue, find the probability that the ball have been selected from the box B.

Solution:

$$P(B|Blue) = \frac{P(B \cap Blue)}{P(Blue)} = \frac{\frac{1}{3} \cdot \frac{19}{34}}{\frac{1}{3} \cdot \frac{15}{23} + \frac{1}{3} \cdot \frac{19}{34} + \frac{1}{3} \cdot \frac{12}{30}} = 0.3468804572$$

- A company has 3 Ford trucks and 2 Man trucks in the region A, 3 Ford trucks and 1 Man trucks in the region B and 2 Ford trucks and 2 Man trucks in the region C. The manager of this company has seen a Ford truck that belongs to her company in a highway which their all trucks equally likely use. What is the probability that the truck belongs to the region B.

Solution:

$$P(B|Ford) = \frac{P(B \cap Ford)}{P(Ford)} = \frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{4}} = \frac{\frac{3}{4}}{\frac{12}{20} + \frac{15}{20} + \frac{10}{20}} = \frac{15}{37}$$

- A shop send monthly an invoice to their customers who do shopping on credit. It is estimated that the probability of a customer's making payment on the date is 0.90 if he/she made payment on the date in the previous month. The same probability is 0.40 if the customer did not make payment on the date in the previous month. What is the probability that
 - a customer will make payment for the next 3 months, given that he/she made payment on the date in this month,
 - a customer who did not make payment on the date in this month will **not** make payment for next two months and then will make payment in the third month?

Solution:

A: a customer -who made payment on the date- will make payment

B: a customer -who did not make payment on the date- will make payment

$$P(A) = 0.90, P(B) = 0.40, P(\bar{A}) = 0.10 \text{ and } P(\bar{B}) = 0.60$$

$$a) P(A) \cdot P(A) \cdot P(A) = 0.90^3$$

$$b) P(\bar{B}) \cdot P(\bar{B}) \cdot P(B) = 0.60 \cdot 0.60 \cdot 0.40 = 0.144$$

5. A box contains 5 white and 2 yellow marbles. If 3 marbles are successively selected without replacing them back, what is the probability that the chosen marbles are formed two white and one yellow marbles?

Solution:

$$P(W - W - Y) = \frac{5}{7} \cdot \frac{4}{6} \cdot \frac{2}{5} = \frac{4}{21},$$

$$P(W - Y - W) = \frac{5}{7} \cdot \frac{2}{6} \cdot \frac{4}{5} = \frac{4}{21},$$

$$P(Y - W - W) = \frac{2}{7} \cdot \frac{5}{6} \cdot \frac{4}{5} = \frac{4}{21}$$

$$P(W - W - Y) + P(W - Y - W) + P(Y - W - W) = 4/7$$

II. Yol:

$$P(2W \text{ and } 1Y) = \frac{C(5,2) \cdot C(2,1)}{C(7,3)} = \frac{10 \cdot 2}{\frac{7 \cdot 6 \cdot 5}{3!}} = \frac{4}{7}$$

6. A bag contains 9 cards which are enumerated from 1 to 9. If 2 cards are randomly selected from the bag, what is the probability that the sum of numbers on the chosen cards is 10?

Solution:

S: 2 cards are randomly selected from 9 cards which are numerated from 1 to 9

A: the sum of numbers on the chosen cards is 10 = {{1,9},{2,8},{3,7},{4,6}}

$$n(S) = C(9,2) = 36$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

7. If a pair of dies are rolled, find the probability that the numbers come up equal or 10 in total.

Solution:

S: the numbers on 2 dies

A: the numbers on 2 dies come up equal or 10 in total =

{{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(4,6),(6,4)}}

$$n(S) = 6 \cdot 6 = 36$$

$$n(A) = 8$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{8}{36} = \frac{2}{9}$$

8. A box contains 2 red and 6 white marbles. If 3 marbles are successively selected without replacing them back, what is the probability that the chosen three marbles are **not** same colored (two of the chosen marbles can be same colored)?

Solution:

$$P(R - R - W) \cdot 3 + P(W - W - R) \cdot 3 = \frac{2}{8} \cdot \frac{1}{7} \cdot \frac{6}{6} \cdot 3 + \frac{6}{8} \cdot \frac{5}{7} \cdot \frac{2}{6} \cdot 3 = \frac{6}{56} + \frac{30}{56} = \frac{9}{14}$$

II. Yol:

$$P(\overline{3W}) = 1 - P(3W) = 1 - \frac{C(6,3)}{C(8,3)} = 1 - \frac{\frac{6 \cdot 5 \cdot 4}{3!}}{\frac{8 \cdot 7 \cdot 6}{3!}} = 1 - \frac{20}{56} = \frac{9}{14}$$

9. A box contains 4 blue, 4 red and 4 white marbles. If 3 marbles are selected, what is the probability that the chosen three marbles are formed at least two red marbles?

Solution:

$$P(R - R - R) + P(R - R - B) \cdot 3 + P(R - R - W) \cdot 3 = \frac{4}{12} \cdot \frac{3}{11} \cdot \left\{ \frac{2}{10} + \frac{4}{10} \cdot 3 + \frac{4}{10} \cdot 3 \right\} = \frac{13}{55}$$

II. Yol:

$$P(2R) + P(3R) = \frac{C(4,2) \cdot C(8,1)}{C(12,3)} + \frac{C(4,3)}{C(12,3)} = \frac{\frac{6 \cdot 8}{3!}}{\frac{12 \cdot 11 \cdot 10}{3!}} + \frac{4}{\frac{12 \cdot 11 \cdot 10}{3!}} = \frac{52}{220} = \frac{13}{55}$$

10. If 2 women and 4 men sit on a couch, what is the probability that two women sit adjacently?

Solution:

S: 6 people sit on a couch

A: two women sit adjacently

$$n(S) = 6!$$

$$n(A) = 5! \cdot 2!$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5! \cdot 2!}{6!} = \frac{1}{3}$$

11. There are 3 red and 4 white balls in a box and 2 red and 5 white balls in a second box. If one ball is selected from each boxes, what is the probability that the chosen balls are same colored?

Solution:

$$P(R - R) + P(W - W) = \frac{3}{7} \cdot \frac{2}{7} + \frac{5}{7} \cdot \frac{4}{7} = \frac{26}{49}$$

II. Yol:

$$P(2R \text{ or } 2W) = \frac{C(3,2)}{C(7,2)} + \frac{C(5,2)}{C(7,2)} = \frac{26}{49}$$



12. A community consists of 12 women and 8 men. 6 of the men and 8 of the women can speak English. If a person is randomly selected from the community, what is the probability that the chosen person can speak English or be woman?

Solution:

A: Person can speak English

B: Person is woman

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{14}{20} + \frac{12}{20} - \frac{8}{20} = \frac{18}{20} = \frac{9}{10}$$

13. For a loaded coin, the probability of showing tail is $\frac{2}{5}$ times of the probability of showing head. Find the probability that the coin shows head if it is rolled.

Solution:

A: The coin shows head

B: The coin shows tail

$$P(A) + P(B) = 1 \quad P(A) = x, \text{ then } P(B) = \frac{2x}{5}$$

$$x + \frac{2x}{5} = 1, \text{ then } x = \frac{5}{7}$$

14. If a die and two coins are rolled together, what is the probability that the die shows an even number and the coins show the same side?

Solution:

A: The die shows an even number = {2, 4, 6}

B: Two coins shows the same side = {TT, HH}

S: Two coins are rolled = {TT, TH, HT, HH}

$$P(A \cap B) = P(A) \cdot P(B) = \frac{3}{6} \cdot \frac{2}{4} = \frac{1}{4}$$

15. 2 marbles are chosen from a box that contains 10 marbles which are red and white. If the probability that the chosen both marbles are red is $\frac{1}{3}$, what is the number of red marbles in the box?

Solution:

A: Both marblers are red

x = Number of red marbles

$$P(A) = \frac{x}{10} \cdot \frac{x-1}{9} = \frac{1}{3}, \quad \text{then } x(x-1) = 30 \text{ and } x = 6$$

16. If a random word of 4 letters is written by using each of the letters {A, E, B, C}, what is the probability that the letters 'A' and 'E' present adjacently?

Solution:

S: 4 letters is written by using each of the letters {A, E, B, C}

A: the letters 'A' and 'E' present adjacently

$$n(S) = 4!$$

$$n(A) = 3! \cdot 2!$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3! \cdot 2!}{4!} = \frac{1}{2}$$

17. Two-digit-numbers that is constructed from the set {0, 1, 3, 4, 6} are written on cards. The numbers on the set can used more than one. If a random card is selected, what is the probability that the number on the card is an odd number?

Solution:

S: Two – digit – numbers that is constructed from the set {0,1,3,4,6}

A: the number on the card is an odd number

$$n(S) = 5 \cdot 5 = 25$$

$$n(A) = 5 \cdot 2 = 10$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{10!}{25} = \frac{2}{5}$$

18. A pair of dies are rolled. If it is known that only one of the dies showed 5, find the probability that the sum of the numbers on the dies is 9 or more.

Solution:

S: A pair of dies are rolled. If it is known that only one of the dies showed 5

A: the sum of the numbers on the dies is 9 or more = {(5,4), (4,5), (5,6)(6,5)}

$$n(S) = C(2,1) \cdot C(5,1) = 10$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{10} = \frac{2}{5}$$